

$$\begin{array}{c}
 \begin{array}{c}
 + \\
 \bullet \\
 \diagup \quad \diagdown \\
 J \quad \quad J' \\
 \diagdown \quad \diagup \\
 \triangle \\
 \diagup \quad \diagdown \\
 k_1 \quad k_2
 \end{array} \\
 \boxed{\{\mathfrak{M} \otimes \mathfrak{N}\}_{\gamma\gamma'}}
 \end{array}
 = \sum_{\gamma''}
 \begin{array}{c}
 + \\
 \bullet \\
 \diagup \quad \diagdown \\
 J \quad \quad J'' \\
 \diagdown \quad \diagup \\
 k_1
 \end{array}
 \boxed{\mathfrak{M}_{\gamma\gamma''}}
 \quad - \quad
 \begin{array}{c}
 + \\
 \circ \\
 \diagup \quad \diagdown \\
 J \quad \quad J' \\
 \diagdown \quad \diagup \\
 \triangle \\
 \diagup \quad \diagdown \\
 k_1 \quad k_2 \\
 - \\
 J''
 \end{array}
 \quad - \quad
 \begin{array}{c}
 + \\
 \bullet \\
 \diagup \quad \diagdown \\
 J'' \quad \quad J' \\
 \diagdown \quad \diagup \\
 k_2
 \end{array}
 \boxed{\mathfrak{N}_{\gamma''\gamma'}}$$

Diagrammatic equation showing the decomposition of a tensor product of two diagrams into a sum of three diagrams.

The left-hand side is a diagram representing the tensor product $\{\mathfrak{M} \otimes \mathfrak{N}\}_{\gamma\gamma'}$. It consists of a large triangle with vertices labeled $+$ at the top and $-$ at the bottom. The left edge is labeled J , the right edge is labeled J' , and the bottom edge is labeled k . Inside the triangle, there is a smaller triangle with vertices labeled $+$ at the top and $-$ at the bottom. The left edge of this inner triangle is labeled k_1 , the right edge is labeled k_2 , and the bottom edge is labeled k .

The right-hand side is a sum of three diagrams, each representing a component of the decomposition. The first diagram is labeled $\mathfrak{M}_{\gamma\gamma''}$ and has vertices labeled $+$ at the top and $-$ at the bottom. The left edge is labeled J , the right edge is labeled J'' , and the bottom edge is labeled k_1 . The second diagram is labeled $\mathfrak{N}_{\gamma''\gamma'}$ and has vertices labeled $+$ at the top and $-$ at the bottom. The left edge is labeled J'' , the right edge is labeled J' , and the bottom edge is labeled k_2 . The third diagram is a triangle with vertices labeled $+$ at the top and $-$ at the bottom. The left edge is labeled J , the right edge is labeled J' , and the bottom edge is labeled J'' . Inside this triangle, there is a smaller triangle with vertices labeled $+$ at the top and $-$ at the bottom. The left edge of this inner triangle is labeled k_1 , the right edge is labeled k_2 , and the bottom edge is labeled k .